

**Reviving the Lost Wilderness: Measuring the Educational
Value of Immersive Technologies in Prehistoric Wildlife
Exploration**

Project ID: 25-26J-417

Project Proposal Report

Insandu W M P S

B.Sc. Special (Hons.) Degree in Information Technology Specializing in
Interactive Media

Department Of Information Technology

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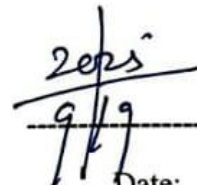
DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the Supervisor:


Date:

Signature of the Co-Supervisor:

Date:

ABSTRACT

This component investigates whether phone-based AR with short knowledge-testing tasks can replace passive label reading and improve learning at the Natural History Museum of Sri Lanka (Osteology Gallery). The system guides young visitors via a treasure-map, uses QR codes at exhibits, presents a one-fact card + a quick quiz, and unlocks a fossil fragment on a correct answer. Collected fragments assemble into a completed skeleton, reinforcing key ideas. The study runs an A/B evaluation: an AR-quiz group versus a labels-only group. We measure engagement (time at exhibits, completion rate) and learning (pre/post scores, in-app accuracy). Content is triaged with curators; analytics are anonymized. Outcomes will show whether simple, phone-only AR can measurably increase engagement and knowledge retention in youth and offer a low-cost, scalable model for Sri Lankan museums.

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1.0 INTRODUCTION

1.1 Background

Museums serve as vital spaces for education, cultural preservation, and public engagement. However, many Sri Lankan museums rely on static displays, labels, and textual descriptions that often fail to inspire younger audiences. Fossil exhibits, while scientifically significant, can be difficult for visitors to interpret without context, limiting their ability to engage and learn effectively.

Augmented Reality (AR) offers a promising way to bridge this gap. Globally, AR has been shown to enrich learning by adding interactive, digital content to physical artifacts, transforming passive visits into participatory experiences [1][9]. Yet, most applications focus on single-user engagement, which isolates visitors instead of encouraging collaboration [3].

By combining AR with gamification and collaboration, museums can introduce treasure-hunt style exploration, quizzes, and shared experiences that improve motivation, curiosity, and knowledge retention. This proposal introduces a QR code-driven AR fossil tour tailored to Sri Lanka, designed to turn static exhibits into immersive, interactive journeys.

1.2 Literature Review

Research on AR in cultural heritage and museum settings shows strong evidence for the potential of multi-user, gamified AR applications to improve visitor experience [3].

Bitter (2022) studied multi-user AR in museums and cultural heritage spaces [2]. The research found that synchronous AR systems, where visitors interact with shared digital objects, encourage collaboration and group reflection. A prototype called “Follow the Blue Butterfly” demonstrated how users with AR devices could follow a virtual guide and work together to annotate skeleton displays in real time. This shows the feasibility and advantages of shared AR learning environments in museums.

Singh et al. (2024) introduced the AREA engine, which combined AR learning applications with emotional engagement analytics. Their study showed that motion data, interaction time, and behavioral responses could be linked to emotions like curiosity, surprise, and frustration. This offers new insights into how learners feel when experiencing AR. It supports the idea of adding quiz-based reflection and emotional checkpoints to AR tours. By doing this, museums can create experiences that are both educational and emotionally engaging.

Prieto Andreu (2025) carried out a systematic review of over 60 studies on collaborative learning in AR, VR, and MR environments [4]. The review revealed that immersive, multi-user experiences consistently boosted social interaction, creativity, and engagement in educational situations. However, issues like high costs, technical limitations, and lack of teacher training still pose challenges for widespread use. These findings emphasize the need for cost-effective and user-friendly solutions, such as a smartphone-based AR tour using QR codes.

Practical applications also demonstrate the potential of AR gamification in education [6]. Systems like Merge EDU and CoSpaces Edu allow learners to interact with 3D AR models together. Museum prototypes like Multi-User AR Experiences in Museums let visitors explore exhibits together through shared AR sessions [6]. Likewise, GeoQuest AR (2023) used scavenger-hunt mechanics to engage children in STEM learning by having them collect virtual fossils and put them into complete models. These examples show that scavenger hunt-style AR tours effectively motivate exploration and reinforce knowledge.

Overall, the literature indicates that while AR is being used more in museums, there is still an opportunity to connect multi-user synchronization, gamified exploration, and educational reflection into a single, cohesive system.

1.3 Research Problem

1.3.1 Main Research Question

“How can immersive technologies be designed and evaluated to transform passive prehistoric-wildlife displays in natural history museums into interactive experiences that measurably increase engagement, knowledge, and empathy?”

1.3.2 Sub Research Questions

1. “Does VR behavior AI for extinct animals create dynamic, non-repetitive interactions and increase visitor engagement and empathy versus scripted behavior on museum-grade hardware?”
2. “Does a real-time visual ecosystem demo with on-the-spot habitat switching raise curiosity and systems understanding versus a non-interactive video, while staying smooth and source-traceable?”
3. “Do QR-code AR tours on visitors’ phones improve wayfinding, engagement, and learning versus a printed-label tour?”

4. “Does phone-based AR quizzes turn fossil exhibits into interactive journeys and outperform labels for youth engagement and learning”

(Covering Point by the Researcher)

1.4 Research Gap

An identifiable gap is present when comparing what this component will bring forward compared to what's currently been implemented. The first and most prevalent factor this component can conduct uniquely is creating evidence to parameter pipeline for extinct behavior. It collectively means that during the development of the AI used to simulate extinct fauna within the VR application the scripting will be built to accept characteristics of those animals and the factors that influence their behaviors into account. The second factor to notice when considering the gap is that most existing VR simulations made for museums often place emphasis on creating a high quality and realistic looking environment rather than focusing on the animals found in that environment [8]. They lack polished or accurate animations and behavior patterns mostly placed into the environment to make it seem alive. Through this research that gap is expected to fill by ensuring that animals exhibit lifelike behaviors and dynamics adapting to their surroundings which leads to the overall atmosphere within the simulation to further be enhanced. Another important factor is that currently no museum VR project has been conducted especially focusing towards transporting the user to a simulation of the prehistoric era of Sri Lanka. This research can mark a pivotal step towards immersive museum experiences in the country, and this component allows all museum goers to observe extinct animals living in the era and environment they belong in [9]. Finally, this component seeks to integrate the user reaction metrics used to measure user responses towards the gameplay in real time to the game's code itself. By tracking the user's behavior through built in head, gaze and proximity trackers to record how the player interacts with the environment, and its animals allow to gain statistical data on user reactions in real time without requiring any physical or external sensors.

2.0 OBJECTIVES

2.1 Main Objective

Design, implement, and evaluate a phone-based, evidence-driven AR quiz tour for fossil exhibits at the Natural History Museum of Sri Lanka. The system should replace static labels with QR-guided tasks and micro-quizzes, run on common smartphones, and show measurable gains in youth engagement and knowledge retention compared with labels alone.

2.2 Specific Objectives

- Develop AI driven behavioral systems that allow animals in the VR environment to act dynamically.
- Ensure optimum code optimization to enable the application to perform smoothly which leads to uninterrupted and enjoyable gameplay experience.
- To evaluate user engagement and reactions through real time testing and feedback mechanisms implemented within the VR experience itself.

3.0 METHODOLOGY

3.1 System Diagram

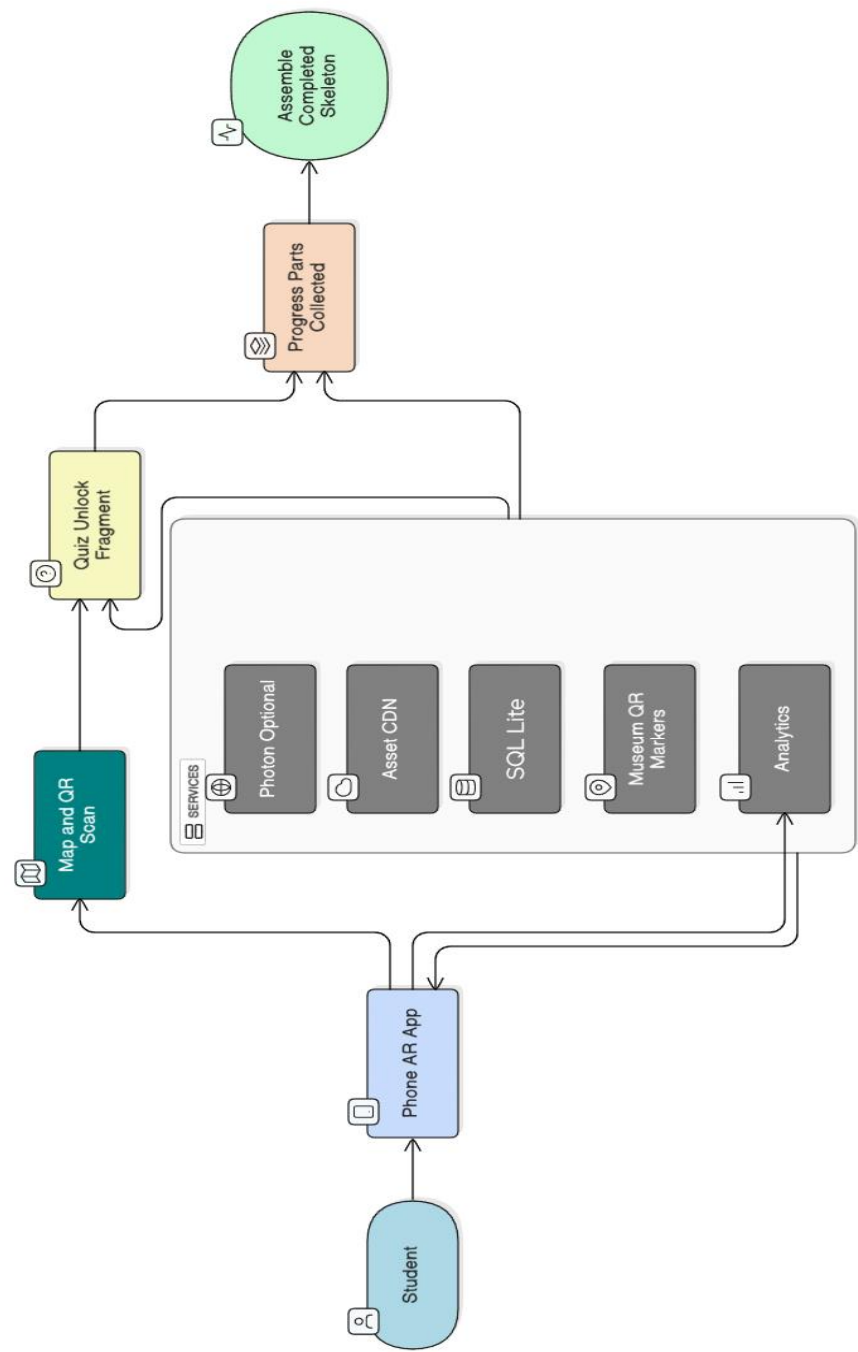


Figure 1: System Architecture Diagram

3.2 Development Procedure

1 Requirement Analysis

- Conduct site visits to the selected museum to study exhibit layouts and identify optimal locations for placing QR codes.
- Interview museum staff to gather input on visitor engagement challenges and content accuracy.
- Review existing AR museum applications to identify best practices and avoid known pitfalls.

2 System Design

- **Architecture:** The system will be divided into modules:
 - QR Code Recognition Module (scans and anchors AR objects).
 - Fossil Fragment Visualization Module (renders 3D fragments).
 - Quiz & Unlock Module (delivers short questions, unlocks items upon correct answers).
 - Skeleton Assembly Module (allows arranging collected fragments into a full skeleton).
 - Multi-User Synchronization Module (ensures shared experience across devices).
- **User Flow:** Visitors start by scanning their first QR code. They collect fragments, solve quizzes, and progress through the museum until they have gathered all required pieces. At the end, the system guides them to assemble and animate the skeleton.

3 Implementation

- **Development Platform:** Unity 3D with AR Foundation to ensure cross-device AR support (ARCore for Android, ARKit for iOS) [10].
- **Networking:** Photon Unity Networking (PUN 2) for real-time synchronization between multiple devices [5].
- **AR Anchoring:** QR codes serve as physical markers, ensuring reliable placement of AR fossils regardless of lighting or space constraints.
- **Content Creation:** Fossil fragments will be modelled or adapted from 3D scans provided by museum data. Animations will be added to bring the assembled skeleton to life.
- **Quiz Integration:** Short multiple-choice or true/false questions will be designed in consultation with museum experts. Correct answers will trigger fragment unlocks, while wrong answers provide hints for learning reinforcement.

4 Testing and Pilot Deployment

- **Lab Testing:** Early versions of the system will be tested in controlled environments to check AR stability, QR recognition accuracy, and network performance.
- **Museum Pilot:** A small-scale trial will be conducted with a sample of visitors. Observations and feedback will be collected regarding usability, engagement, and educational value.

5 Evaluation

- **Quantitative Data:** Number of fragments collected, quiz success rates, time spent, and completion rates.
- **Qualitative Data:** Visitor interviews, surveys, and observation notes regarding enjoyment, motivation, and learning.
- **Analysis:** Evaluate whether the scavenger hunt AR system improves user experience compared to static exhibits[8].

6 Iteration and Refinement

- Based on pilot results, refine quiz design, user interface, and synchronization performance.
- Make improvements to ensure scalability so the system can be expanded to other exhibits and museums.

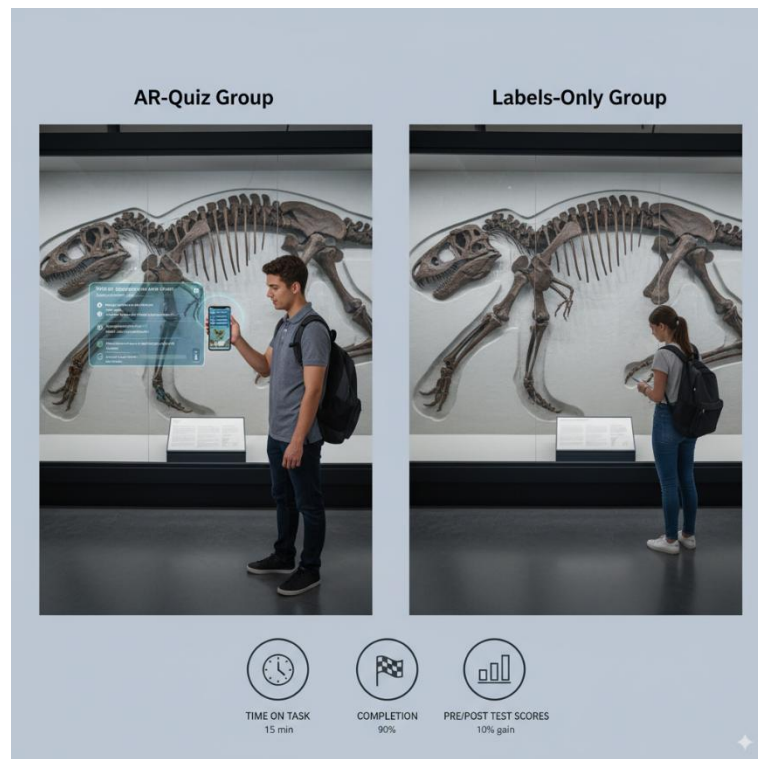


Figure 2: Split-screen A/B evaluation

Tools and Technologies

- **Unity 3D + AR Foundation** – Cross-platform AR development.
 - **Photon Unity Networking / Unity Netcode** – Multi-user session management and synchronization [5].
 - **Blender / 3ds Max** – 3D modelling of fossil fragments and skeletons.
 - **QR Code Library (Zxing or AR Foundation marker support)** – Recognition and anchoring of AR objects.
 - **Firebase / Local Database** – (Optional) Storing quiz questions, progress, and user session data.
 - **Android and iOS Smartphones/Tablets** – Deployment devices for testing and museum use.
-

Deliverables

- A functional **AR mobile application** enabling QR-based fossil scavenger hunts.
- A **multi-user synchronized session** feature for collaborative tours.
- A set of **interactive quizzes** integrated into the exploration process.
- A **skeleton assembly and animation** module providing an engaging finale.
- **Evaluation results** demonstrating improvements in user experience and educational outcomes.

3.3 Gantt Chart

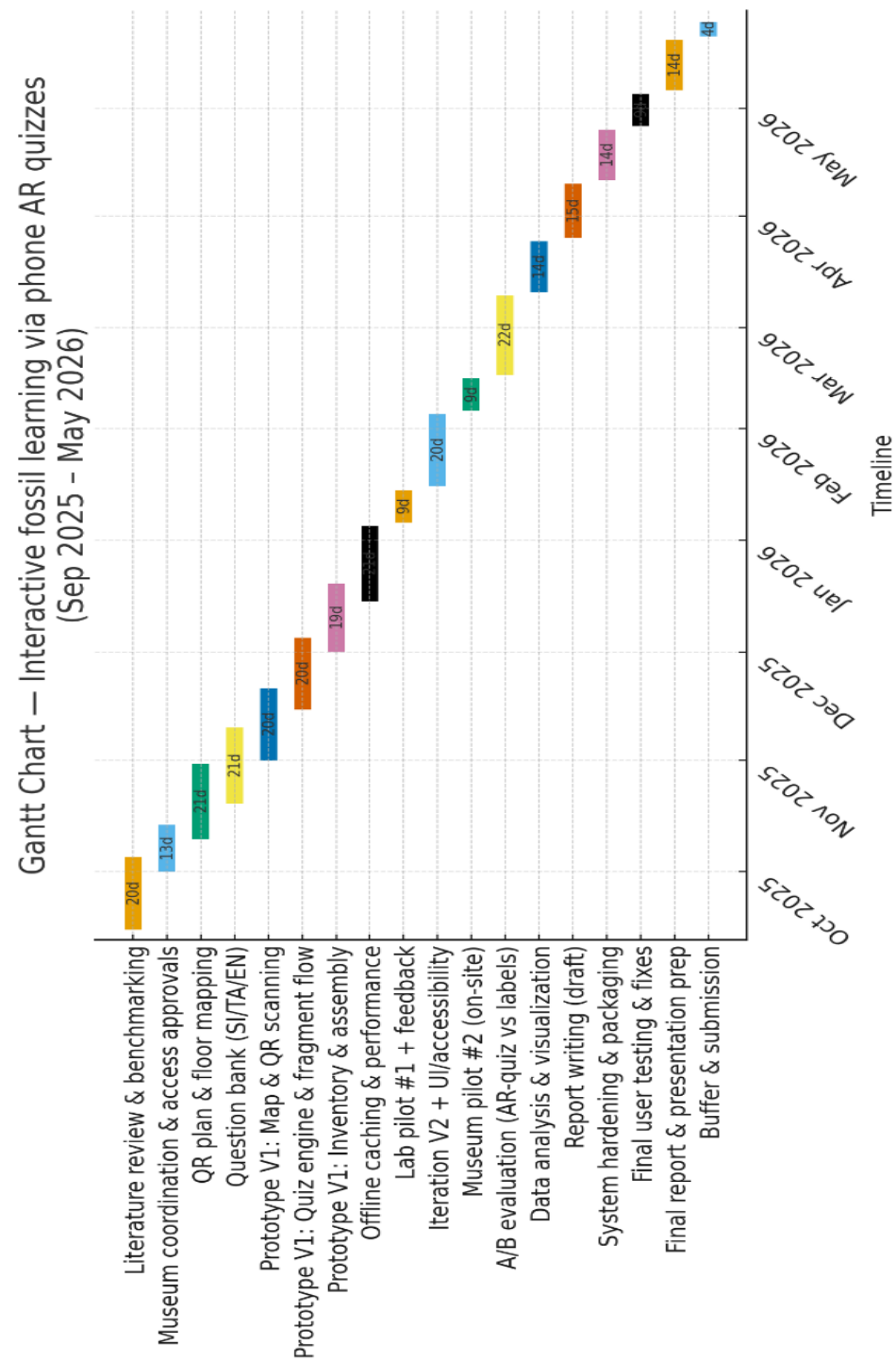


Figure 3: Project Gantt Chart



Figure 4: Setting & participants (photo)

4.0 PROJECT REQUIREMENTS

4.1 Functional Requirements

1. QR Code Anchoring

The system must recognize QR codes placed near fossil exhibits and use them as anchors to render AR fossil fragments. Each scanned code should trigger the visualization of a specific fragment tied to the corresponding exhibit..

2. Virtual Fossil Fragment Visualization

Each QR code should reveal a **3D model of a fossil fragment** that can be viewed in AR. Visitors must be able to rotate, zoom, and examine the fragment in detail.

3. Gamified Collection Mechanism

The system must allow visitors to **collect virtual fragments** into their personal inventory within the app. This ensures progression through the scavenger hunt and motivates visitors to explore the museum further.

4. Quiz-Based Unlocking

Certain fossil fragments should be “locked” behind quiz challenges. Visitors must correctly answer **short educational questions** related to the exhibit to unlock these items. This ensures active engagement with exhibit content and promotes retention.

5. Skeleton Assembly Module

After collecting all fragments, the system must allow visitors to assemble the pieces into a complete skeleton. The app should provide guidance to ensure correct placement of each bone and offer feedback during the assembly process.

6. AR Animation of Completed Skeleton

Once the skeleton is fully assembled, the system must animate the model to show the creature’s movement or behaviour, providing a memorable and rewarding finale to the scavenger hunt.

7. Multi-User Synchronization

The application must support shared sessions, where multiple visitors can participate in the same scavenger hunt. Progress such as collected

fragments or skeleton assembly should be visible to all participants in real time.

8. User Progress Tracking

The system must track each visitor's progress through the scavenger hunt, including the number of fragments collected, quizzes attempted, and skeleton completion status.

4.2 Non-Functional Requirements

1. Performance

- The system must maintain smooth AR rendering at a minimum of 30 frames per second (FPS). QR recognition should occur in less than 2 seconds to ensure fluid interaction.

2. Scalability

- The system should be designed in a modular way so that additional exhibits, fossil fragments, or quiz questions can be added easily without reworking the entire app.

3. Usability

- The interface must be simple and intuitive, requiring minimal instructions. Clear visual cues and icons should guide visitors through scanning, collecting, and assembling tasks.

4. Reliability

- The AR visualization must remain stable even in varying lighting conditions within the museum. QR code markers must be robustly recognized even when partially obscured or scanned at an angle.

5. Accessibility

- The system should be accessible to a wide age range of visitors, including school children. Text size, color contrasts, and simple navigation must be optimized for readability and inclusivity.

6. Security & Privacy

- Visitor progress and quiz responses must be handled securely. Any collected data (such as usage statistics for evaluation purposes) must be anonymized to protect privacy.

7. Maintainability

- The system must follow modular architecture, allowing updates to AR content, quizzes, and analytics independently.

8. Cost-Efficiency

- The system should run entirely on consumer smartphones/tablets without requiring specialized AR hardware.

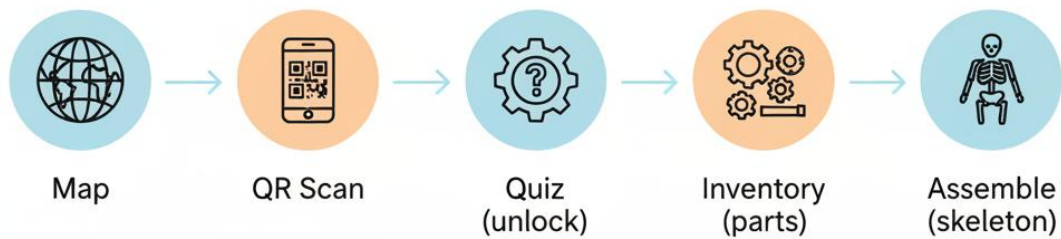


Figure 5: Overview

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